

Evangelis Silva

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Research Interests

Database and Distributed Systems (Detailed Topics: Graph Database Systems, Storage Architectures, Indexing Strategies, and Real-time Data Processing)

Education

University at Albany, SUNY, Ph.D. in Computer Science Jan 2022 – Present

- Advisor: Prof. Jeong-Hyon Hwang
- Related Coursework: Database Systems, Data Structures & Algorithms, Software Engineering, Data Mining, Formal Languages and Theory of Computation, Computer Communications Networks

University of Moratuwa, Sri Lanka, BS.c. (Hons.) in Information Technology & Management Jan 2014 – Mar 2019

- GPA: 3.50/4.20
- Advisor: Dr. Supunmali Ahangama
- Related Coursework: Advanced Database Management Systems, Data Mining & Data Warehousing, Big Data Analytics, Essentials of Computer Organization & Architecture, Software Engineering Methods, Artificial Intelligence, Enterprise Application Development

Experience

Graduate Assistant, University at Albany, SUNY Jan 2022 – Present

- Conduct research in databases and distributed systems, focusing on the design and optimization of high-throughput on-disk and in-memory key-value stores.
- Collaborate with faculty on system-level experiments, distributed benchmarking, and publications.
- Assist in teaching undergraduate and graduate computer science courses, supporting labs and assessments.

Senior Software Engineer, Sysco LABS, Sri Lanka Nov 2020 – Dec 2021

- Designed and scaled microservice architectures for Sysco's high-volume enterprise systems.
- Built RESTful APIs and distributed data pipelines serving tens of thousands of users.
- Improved query performance by 75% through optimized indexing and query restructuring.
- Enabled near real-time data sync between monolithic and microservice databases using CDC and data streaming.
- Partnered with cross-functional teams to enhance scalability, resilience, and observability.

Software Engineer, Sysco LABS, Sri Lanka Mar 2019 – Nov 2020

- Developed and maintained RESTful APIs and backend services for Sysco's global digital platform.
- Implemented CI/CD pipelines using Jenkins and Concourse, improving deployment reliability.
- Built observability pipelines with Datadog and Kibana for proactive monitoring and alerting.
- Enhanced data traceability through audit logging and event tracking across microservices.
- Collaborated with globally distributed teams on system design, code reviews, and production reliability.

Engineering Intern (Data Analytics), Vesess, Sri Lanka Oct 2017 - Mar 2018

- Developed web scraping and data enrichment workflows for the Donordo analytics platform.
- Performed exploratory data analysis and built Python-based data processing pipelines.
- Created data visualizations to support donor-matching insights.

- Assisted in cloud deployment and caching optimizations to improve responsiveness.

Selected Projects

An Experimental Evaluation and Investigation of Group Farness Estimation Techniques

- Conducted a comprehensive experimental study on scalable group farness estimation in large-scale graphs, extending single-source farness algorithms to the group setting.
- Evaluated sampling-based and cardinality estimation methods, including KMV and Cohen's HIP estimator, analyzing trade-offs among accuracy, memory, and runtime efficiency.
- Demonstrated that Cohen's method consistently achieves 2–5% accuracy gains and improved scalability across diverse graph structures.
- **Tools and Libraries:** Java, Bash, Python (NumPy, Matplotlib, Pandas)

Mimix Data Transformer

Demo: Watch on YouTube

- Developed a modular Java library for efficient and extensible processing of Mimix event streams, enabling transformation, mapping, and post-processing workflows for relational and streaming data systems.
- Implemented fine-grained event filtering, JSON-based mapping, and custom transformation hooks for dynamic data workflows.
- Integrated with AWS Kinesis for real-time event streaming and replication to downstream systems.
- Designed a pluggable architecture supporting custom filters, transformers, and database writers, packaged as a Maven-compatible artifact for enterprise integration.
- **Tools and Libraries:** Java, Spring Boot, PostgreSQL, AWS Kinesis, Amazon EC2, Amazon SNS, Amazon S3, Maven, JSON, Jenkins

Enterprise Email Service

- Designed and developed a robust internal email service platform for enterprise systems at Sysco LABS, enabling automated and template-based email communication across multiple business products.
- Architected scalable RESTful APIs and database schemas to support high-volume email workflows and seamless integration with existing enterprise applications.
- **Tools and Libraries:** Java, Spring Boot, Amazon SNS, Amazon SQS, Amazon EventBridge, Amazon ECS, AWS Lambda, PostgreSQL, Concourse, Jenkins, Datadog, Git, Jira

Compassionate Sri Lanka

- A CSR initiative at Sysco LABS with the Ministry of Social Empowerment to build a responsive web platform connecting donors with individuals in need.
- Developed a responsive front-end and RESTful APIs enabling donor–recipient interaction, request tracking, and data management.
- Integrated SMS and email notifications for real-time communication and status updates.
- Enhanced the platform's scalability and accessibility for nationwide deployment.
- **Tools and Libraries:** Node.js, AngularJS, Amazon SES, MongoDB, Git

Publications

E. Silva, M. Nandathilaka, S. Dalugoda, T. Amarasinghe, S. Ahangama, and G. T. Weerasuriya. "Machine Learning-Based Automated Tool to Detect Sinhala Hate Speech in Images." *Proceedings of the 6th International Conference on Information Technology Research (ICITR 2021)*, IEEE.

Skills

Programming, Scripting, and Query Languages: Java, Python, C++, Rust, JavaScript, SQL, Bash

Frameworks & Libraries: Spring Boot, Node.js, AngularJS, NumPy, Pandas, Matplotlib

Cloud & Infrastructure: AWS (EC2, S3, SNS, SQS, SES, Lambda, EventBridge, Kinesis, ECS), Docker, Jenkins, Concourse CI

Databases: PostgreSQL, MongoDB, RocksDB, Amazon DynamoDB, Redis

Data & Systems Tools: Datadog, Kibana, Maven, Gradle, Git, Jira, Elasticsearch

Research & Analytical Tools: Python (NumPy, Matplotlib, Pandas, scikit-learn, NLTK), LaTeX, Experimental Evaluation, Benchmarking, Data Visualization

Soft Skills: Technical Documentation, System Design, Cross-functional Collaboration, Teaching Assistance, Research Writing

Awards & Achievements

- **Best Paper Award (AI & ML Track)** — awarded at the 6th International Conference on Information Technology Research (ICITR 2021), IEEE. *Dec 2021*. for the paper titled, Machine Learning-Based Automated Tool to Detect Sinhala Hate Speech in Images
- **Sysco LABS Customer First Value Award 2020** — awarded for generating business value while effectively solving and serving customer needs in the Digital Vertical *Dec 2020*.
- **Dean's List Appearance** — achieved Dean's List recognition in Level 2, Semester 2 for outstanding academic performance at University of Moratuwa, Sri Lanka

Professional Service & Leadership

- **Conference Paper Reviewer** — 36th International Conference on Scientific and Statistical Database Management (SSDBM 2024)
- **Student Relations Lead** — Vesess Alpha Coding Challenge (www.vesess.com/alpha)
- **Treasurer** — Sysco LABS SL Toastmasters Club
- **Editor** — *FIT Chronicles (2017)*, Faculty of IT Newsletter, University of Moratuwa
- **Technical Coordinator & Article Reviewer** — fit2school, Faculty CSR Project
- **Organizing Committee Vice President (Delegate)** — IT Recruitment 2016, AIESEC Colombo South
- **Organizing Committee Vice President (Finance)** — Global Classroom 1.0, AIESEC Colombo South
- **Organizing Committee Member** — Heal a Nation 4.0, AIESEC Colombo South
- **Technical Session Conductor (O/L Outreach)** — INTECS Outreach, Kurunegala & Ruwanwella
- **Facilitator** — Catholic Students' Movement, University of Moratuwa
- **Hardware Project Demonstrator** — Faculty of IT, Future Minds 2015 Exhibition